LIBRARY MANAGEMENT SYSTEM

Project Documentation & Operational README

Domain: SQL DB Management
Engine: MySQL 8+

1. PROJECT DESCRIPTION

The **Library Management System** is a relational database solution developed using SQL. It centralizes member information, tracks the complete book catalog, manages borrow and return transactions, maintains stock availability, and automatically calculates overdue fines.

2. TECHNOLOGIES USED

MySQL 8+ | **SQL Standard** | **Primary & Foreign Keys** | **SQL JOIN Queries** | **Aggregate Functions** | **CASE Expressions**

3. DATABASE SCHEMA TABLES

Books Table

- **BookID** (PK)
- Title
- Author
- Category
- AvailableCopies

Members Table

- **MemberID** (PK)
- MemberName
- Email
- JoinDate

Transactions Table

- **TransactionID** (PK)
- BookID (FK)
- MemberID (FK)
- BorrowDate / DueDate
- ReturnDate / Status

4. KEY SYSTEM FEATURES

- Stores & maintains full book catalog information
- Manages library member profiles & registrations
- Tracks borrow, return, and pending status
- Calculates daily overdue fines based on schedules
- Monitors real-time inventory and copy availability
- Generates lists of overdue items & active borrowings

5. FINE ASSESSMENT SCALE

OVERDUE DURATION	DAILY FINE RATE	STATUS / IMPACT
1–3 Days	$1.00 / day	Standard Reminder
4–7 Days	$2.00 / day	Notice Issued
8–14 Days	$3.00 / day	Account On Hold
Above 14 Days	$5.00 / day	Membership Suspended

6. EXECUTION & SETUP INSTRUCTIONS

1. Open MySQL Workbench or your preferred MySQL client software.
2. Run the SQL script to create the database and table schemas.
3. Execute the sample data insertion script to populate test records.
4. Run queries to view book availability, active borrowings, and fines.